DSTALGO
First Activity for Final Period:

1. Problem: Find the largest number in a two-dimensional array.
2. Problem: Ask the user to enter six numbers and display them in a table containing two rows and three columns.
3. Problem: A teacher wants to calculate the average grade of each student across three subjects.
4. Problem: Create a 3 × 3 multiplication table using a two-dimensional array.

Strict submission deadline: late work will not be considered

Write the program and output below:

1.
```
//find the largest number in a two dimensional array
int[,] numbers = new int[,] { { 5, 10, 15}, { 2, 4, 6 }, { 3, 6, 9 } };
int largest = numbers[0, 0];
for (int i = 0; i < numbers.GetLength(0); i++)
{
    for (int j = 0; j < numbers.GetLength(1); j++)
    {
        if (numbers[i, j] > largest)
        {
            largest = numbers[i, j];
        }
    }
}
Console.WriteLine("The largest number is: " + largest);
```



2.
```
//Ask the user to enter six numbers and display them in a table cotaining two rows and three columns
int[] num = new int[6];
for (int i = 0; i < num.Length; i++)
{
    Console.Write("Enter number " + (i + 1) + ": ");
    num[i] = Convert.ToInt32(Console.ReadLine());
}
Console.WriteLine("The numbers are:");
```

```
for (int i = 0; i < 2; i++)
{
    for (int j = 0; j < 3; j++)
    {
        Console.Write(num[i * 3 + j] + " ");
    }
    Console.WriteLine();
}
```

```
Enter number 1: 5
Enter number 2: 6
Enter number 3: 7
Enter number 4: 8
Enter number 5: 9
Enter number 6: 10
The numbers are:
5 6 7
8 9 10
```

3.
```
//A teacher wants to calculate the average grade of each student across three subjects.
int[] grade = new int[3];
for (int i = 0; i < grade.Length; i++)
{
    Console.Write("Enter grade for subject " + (i + 1) + ": ");
    grade[i] = Convert.ToInt32(Console.ReadLine());
}
double average = grade.Average();
Console.WriteLine("The average grade is: " + average);
```

```
Enter grade for subject 1: 55
Enter grade for subject 2: 60
Enter grade for subject 3: 88
The average grade is: 67.66666666666667
```

4.
```
//Create a 3x3 multiplication table using two dimensional array
int[,] multiply = new int[5, 5];
for (int i = 0; i < multiply.GetLength(0); i++)
{
    for (int j = 0; j < multiply.GetLength(1); j++)
    {
        multiply[i, j] = (i + 1) * (j + 1);
    }
}
Console.WriteLine("The multiplication table is:");
for (int i = 0; i < multiply.GetLength(0); i++)
{
    for (int j = 0; j < multiply.GetLength(1); j++)
    {
```

```
            Console.Write(multiply[i, j] + " ");
        }
        Console.WriteLine();
    }
```

